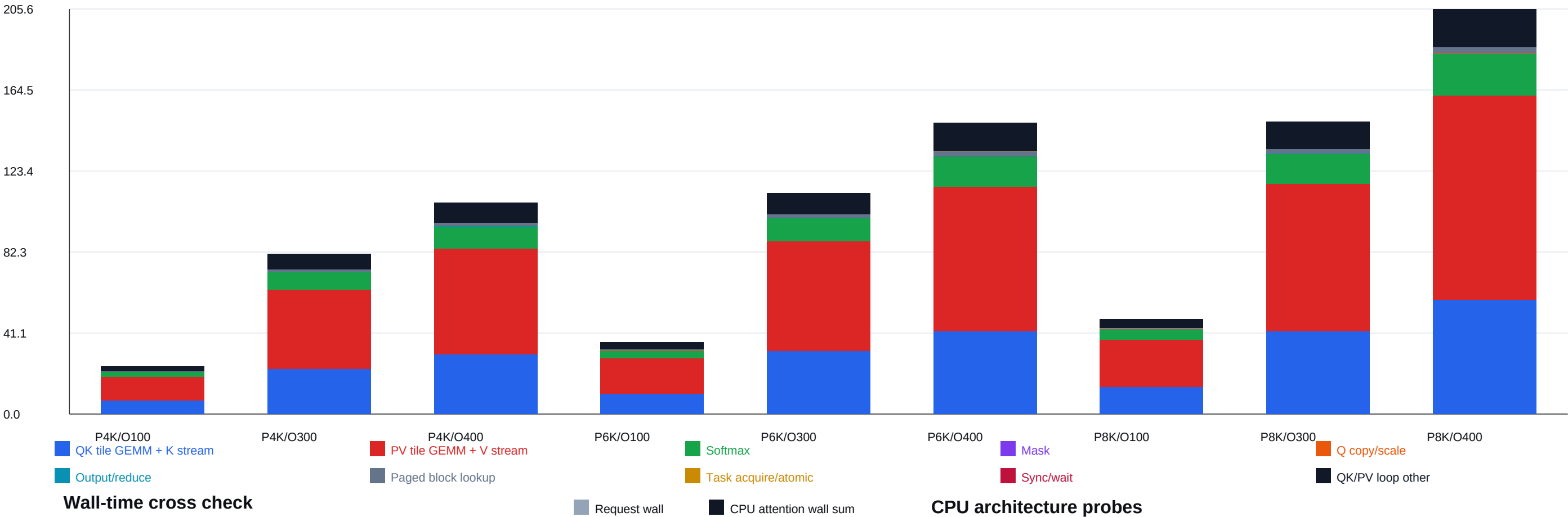


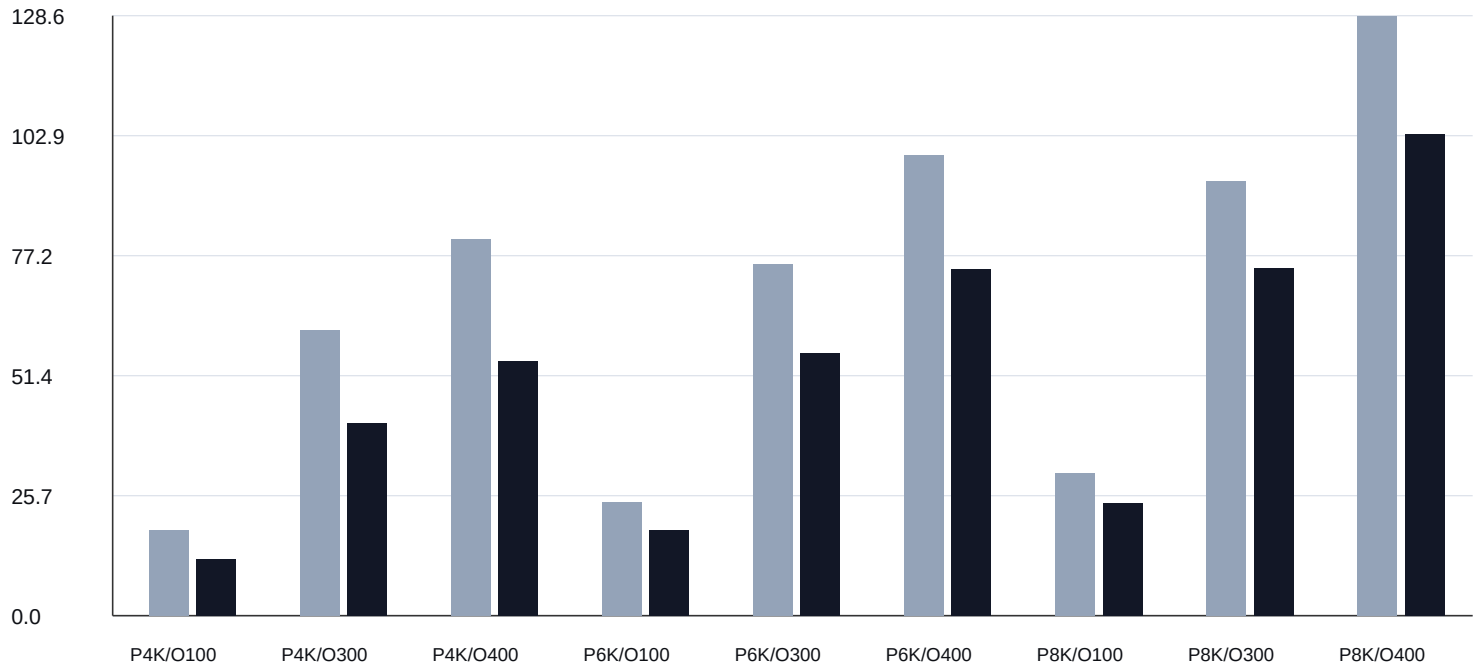
CPU Attention Internal Profile for Heterogeneous Decode

Native vLLM C++ CPU attention; batch=1; PDF generated by Python

Internal thread-time breakdown



Wall-time cross check



CPU architecture probes

Metric	Measured value
OpenMP threads	128
Sequential read bandwidth	80.72 GiB/s
Random pointer chase latency	50.49 ns/access
OpenMP barrier latency	46.70 us/barrier
Contended atomic fetch_add	66.46 ns/op

QK/PV tile GEMM includes arithmetic and K/V cache streaming loads.

Thread-time bars are accumulated over OpenMP workers and exceed wall time.